

Computer Science & IT

Discrete Mathematics



Set Theory & Algebra

Lecture No. 13



By- Vishal Sir



Recap of Previous Lecture



Topic

Bijjective function and inverse of a function

Topic

Identity function and constant function

Topic

Identical function

Topic

Function composition

Topics to be Covered



Topic

Practice questions & PYQ



Consider the set $S = \{1, w, w^2\}$, where w and w^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $\{S, *\}$ forms



MCQ [2010: 1 Mark]

- ☒ A a group
- ☐ B a ring
- ☐ C an integral domain
- ☐ D a field

How many onto (or surjective) functions are there from an n -element ($n \geq 2$) set to a 2-element set?

MCQ [2012: 2 Marks]

- ☐ A 2^n
- ☐ B $2^n - 1$
- ☒ C $2^n - 2$
- ☐ D $2(2^n - 2)$

A binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is TRUE about $\oplus$?



MCQ [2013: 1 Mark]

☒ A

Commutative but not associative

☐ B

Both commutative and associative

☐ C

Associative but not commutative

☐ D

Neither commutative nor associative

$$(x \oplus y) \oplus z$$

$$x \oplus (y \oplus z)$$

$$(x^2 + y^2) \oplus z$$

$$x \oplus (y^2 + z^2)$$

$$(x^2 + y^2)^2 + z^2$$

$$x^2 + (y^2 + z^2)^2$$

$$(1 + 1)^2 + 100^2$$

$$\neq 1 + (1 + 10000)^2$$

Let S denote the set of all functions $f: \{0,1\}^4 \rightarrow \{0,1\}$. Denote by N the number of functions from S to the set $\{0,1\}$. The value of $\log_2 \log_2 N$ is 16.



NAT [2014 (Set-1): 2 Marks]

$$|S| = \text{No. of functions from } \{0,1\}^4 \rightarrow \{0,1\}$$

$\downarrow \qquad \qquad \qquad \downarrow$

$$|\{0,1\}^4| = 16 \qquad |\{0,1\}| = 2$$
$$= 2^{16}$$

$$N : S \rightarrow \{0,1\}$$

$$\text{No. of function}(N) = 2^{|S|}$$

$$N = 2^{(2^{16})}$$

$$\begin{aligned} \log_2(\log_2 N) &= \log_2(\log_2(2^{(2^{16})})) \\ &= \log_2(2^{16}) = \underline{16} \end{aligned}$$

Consider the following relation on subsets of the set S of integers between 1 and 2014. For two distinct subsets U and V of S we say $U < V$ if the minimum element in the symmetric difference of the two sets is in U .

Consider the following two statements:

S_1 : There is a subset of S that is larger than every other subset.

S_2 : There is a subset of S that is smaller than every other subset.

Which one of the following is CORRECT?

MCQ [2014 (Set-2): 2 Marks]

- ☒ A Both S_1 and S_2 are true
- ☐ B S_1 is true and S_2 is false
- ☐ C S_1 is false and S_2 is true
- ☐ D Neither S_1 nor S_2 is true

$$S: \{1, 2, \dots, 2014\}$$

$$U < V \quad \text{iff} \quad \text{Min}(U \Delta V) \in U$$

$$\{4, 6, 8\} \Delta \{1, 4, 9\} = \{1, 6, 8, 9\}$$

$$\text{Min}\{1, 6, 8, 9\} = 1 \in V$$

$$\therefore V \in U$$

$$\{4, 6, 8\} \Delta \emptyset = \{4, 6, 8\} \Rightarrow \text{Min}\{4, 6, 8\} \in \{4, 6, 8\}$$

$$\therefore \{4, 6, 8\} < \emptyset$$

$$\{4, 6, 8\} \Delta S = \{1, 2, 3, 5, 7, 9, \dots, 2014\}$$

$$S < \{4, 6, 8\}$$

Consider the set of all functions $f: \{0, 1, \dots, 2014\} \rightarrow \{0, 1, \dots, 2014\}$ such that $f(f(i)) = i$, for all $0 \leq i \leq 2014$. Consider the following statements:

☒ ~~P~~: For each such function it must be the case that for every i , $f(i) = i$.

☒ ~~Q~~: For each such function it must be the case that for some i , $f(i) = i$.

☒ ~~R~~: Each such function must be onto.

Which one of the following is CORRECT?

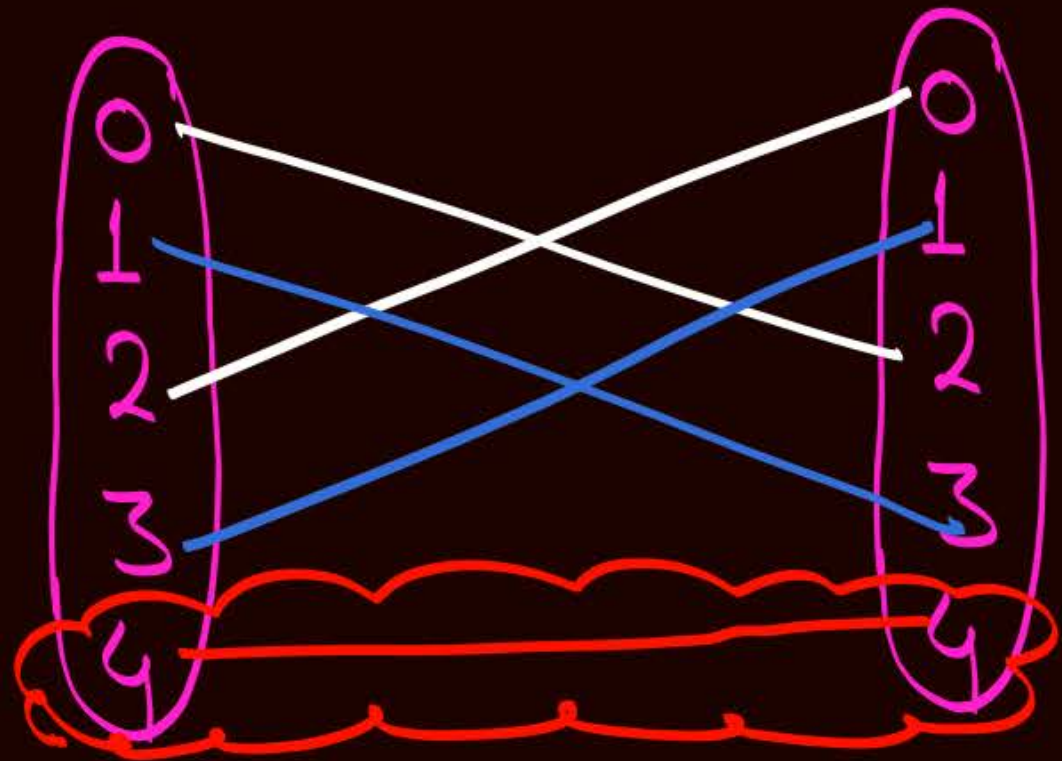
MCQ [2014 (Set-3):2 Marks]

☐ A P, Q and R are true

☒ B Only Q and R are true

☐ C Only P and Q are true

☐ D Only R is true



$$f(f(0)) = f(2) = 0$$

$$f(f(2)) = f(0) = 2$$

If $g(x) = 1 - x$ and $h(x) = \frac{x}{x-1}$, then $\frac{g(h(x))}{h(g(x))}$ is

MCQ [2015 (Set-1): 1 Mark]

- ☒ A $\frac{h(x)}{g(x)}$
- ☐ B $-\frac{1}{x}$
- ☐ C $\frac{g(x)}{h(x)}$
- ☐ D $\frac{x}{(1-x)^2}$

$$\frac{g(h(x))}{h(g(x))} = \frac{g\left(\frac{x}{x-1}\right)}{h(1-x)} = \frac{\left(1 - \frac{x}{x-1}\right)}{\left(\frac{1-x}{(1-x)-1}\right)} = \frac{\cancel{x-1-x}}{\frac{x-1}{(1-x)}} = \frac{-x}{(1-x)}$$

$$= \frac{x}{(x-1)(1-x)} = \frac{h(x)}{g(x)}$$

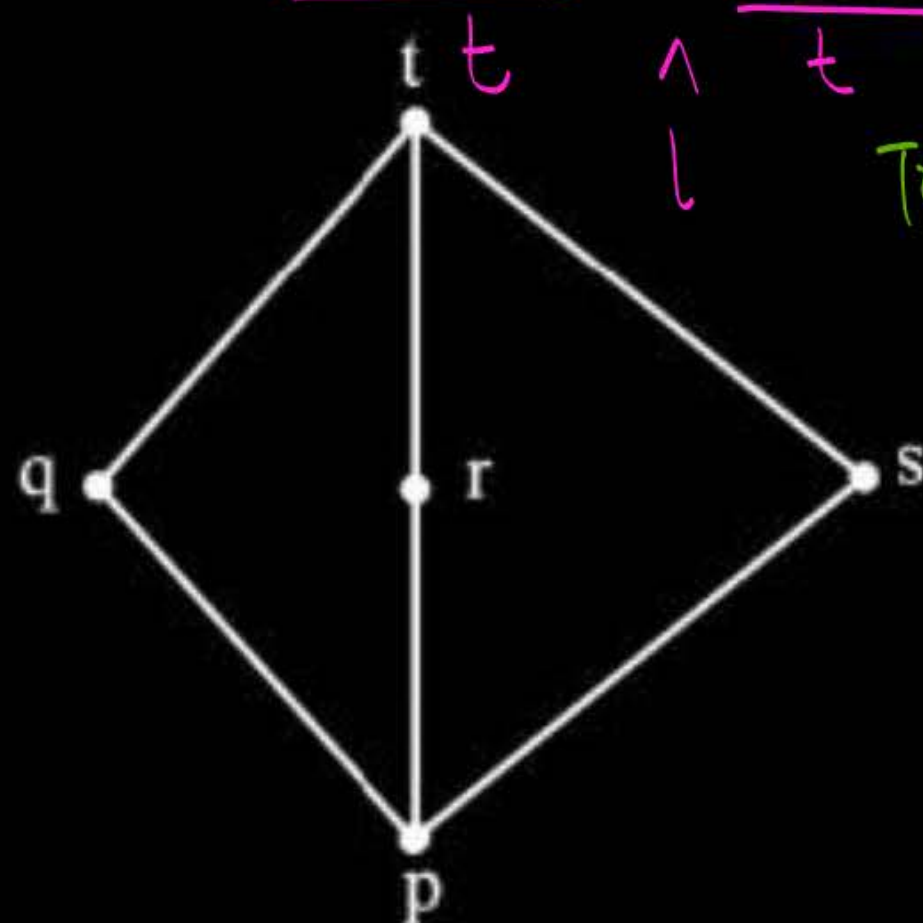


Suppose $L = \{p, q, r, s, t\}$ is a lattice represented by the following Hasse diagram:

For any $x, y \in L$, not necessarily distinct, $x \vee y$ and $x \wedge y$ are join and meet of x, y respectively. Let $L^3 = \{(x, y, z) : x, y, z \in L\}$ be the set of all ordered triplets of the elements of L . Let p_r be the probability that an element $(x, y, z) \in L^3$ chosen equiprobably satisfies $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z)$. Then

MCCQ [2015 (Set-1): 1 Mark]

- ☐ A $p_r = 0$
- ☐ B $p_r = 1$
- ☐ C $0 < p_r \leq \frac{1}{5}$
- ☒ D $\frac{1}{5} < p_r < 1$



Total # order triple (x, y, z)
 $(5 \times 5 \times 5) = 125$

(x, y, z)
 $= t$
 $1 \times 5 \times 5 = 25$



The number of onto functions (surjective functions) from set $X = \{1, 2, 3, 4\}$ to set $Y = \{a, b, c\}$ is_____.

NAT [2015 (Set-2): 2 Marks]

$$\text{Ans} = {}^4C_2 \times 3!$$

$$= \underline{\underline{36}}$$



Let X and Y denote the sets containing 2 and 20 distinct objects respectively and F denote the set of all possible functions defined from X to Y . Let f be randomly chosen from F . The probability of f being one-to-one is _____.

NAT [2015 (Set-2): 2 Marks]

$|F| = 20^2 = 400$

In ' F ' no. of functions that are one-one = 20×19

∴ Probability that ' f ' is one-one = $\frac{20 \times 19}{20 \times 20} = \frac{19}{20} = 0.95$



Let # be a binary operator defined as $X \# Y = X' + Y'$ where X and Y are Boolean variables.

Consider the following two statements:

$$S_1: (P \# Q) \# R = P \# (Q \# R)$$

$$S_2: Q \# R = (R \# Q)$$

Which of the following is/are true for the Boolean variables P, Q and R?

MCQ [2015 (Set-3): 1 Mark]

- ☐ A Only S_1 is TRUE
- ☒ B Only S_2 is TRUE
- ☐ C Both S_1 and S_2 are TRUE
- ☐ D Neither S_1 nor S_2 are True

$$(P' + Q')' + Z'$$
$$PQ + Z'$$

$$P' + (Q' + Z')'$$
$$P' + QZ$$



A function $f: \mathbb{N}^+ \rightarrow \mathbb{N}^+$, defined on the set of positive integers $\mathbb{N}^+$, satisfies the following properties

$$f(n) = f(n/2) \quad \text{if } n \text{ is even}$$

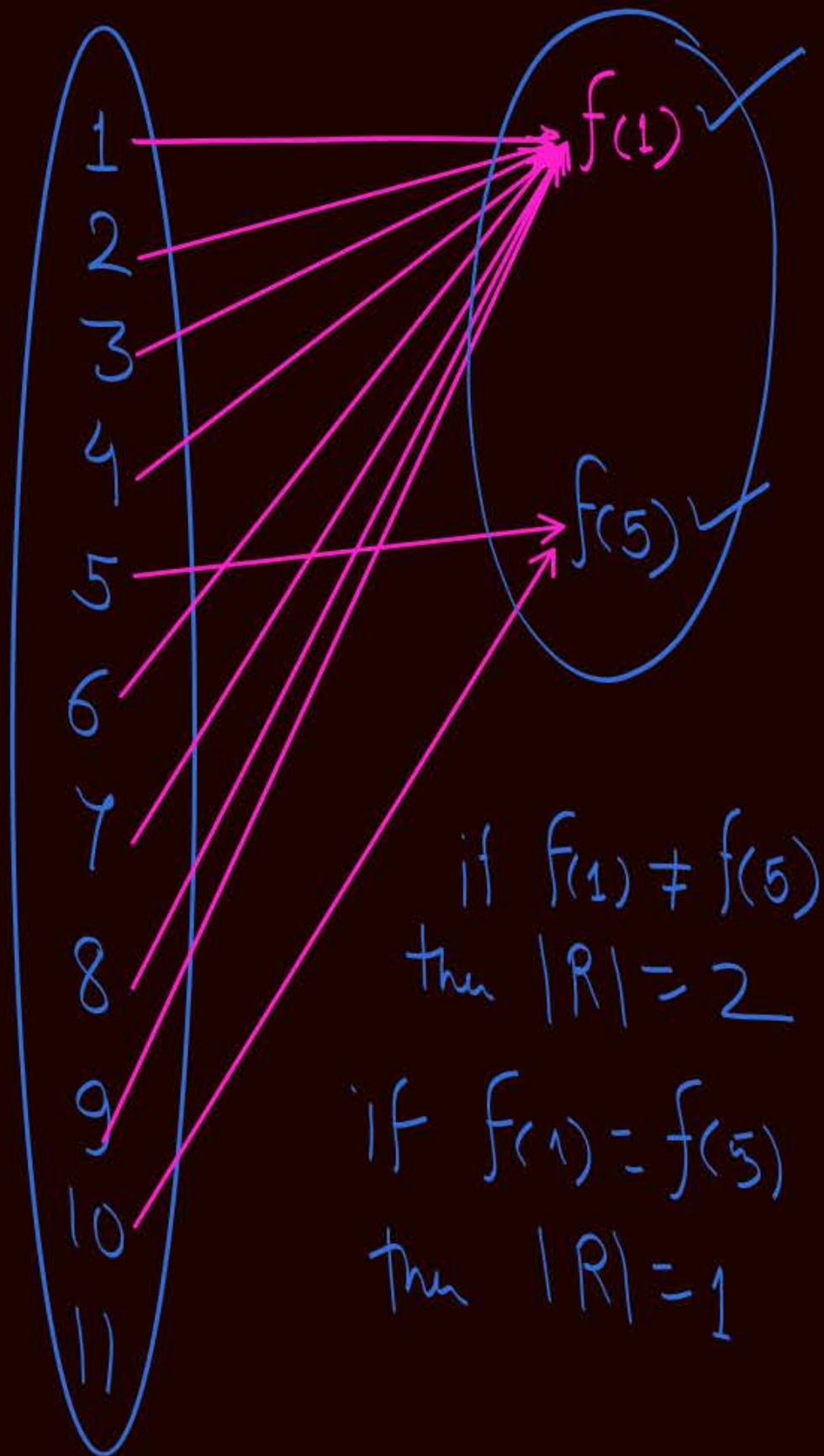
$$f(n) = f(n+5) \quad \text{if } n \text{ is odd}$$

"Range of f"

Let $R = \{i | \exists j: f(j) = i\}$ be the set of distinct values that f takes.

The maximum possible size of R is. 2.

NAT [2016 (Set-1): 2 Marks]



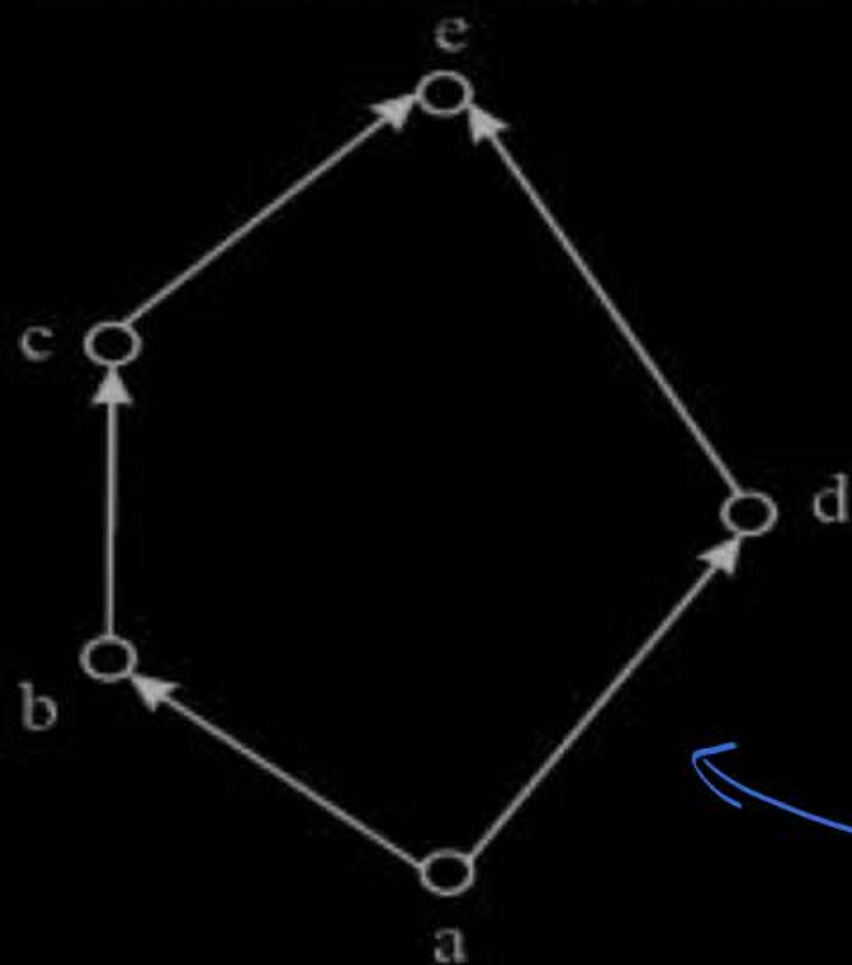
$$f(1) = f(6) = f(3) = f(8) = f(4) = f(2) = f(7) = f(12)$$

$$f(5) = f(10)$$

Consider the set $X = \{a, b, c, d, e\}$ under the partial ordering:

$R = \{(a, a), (a, b), (a, c), (a, d), (a, e), (b, b), (b, c), (b, e), (c, c), (c, e), (d, d), (d, e), (e, e)\}$.

The Hasse diagram of the partial order (X, R) is shown below:



The minimum number of ordered pairs that need to be added to R to make (X, R) a lattice is 0.

NAT [2017 (Set-2): 1 Mark]

← it is already L^*

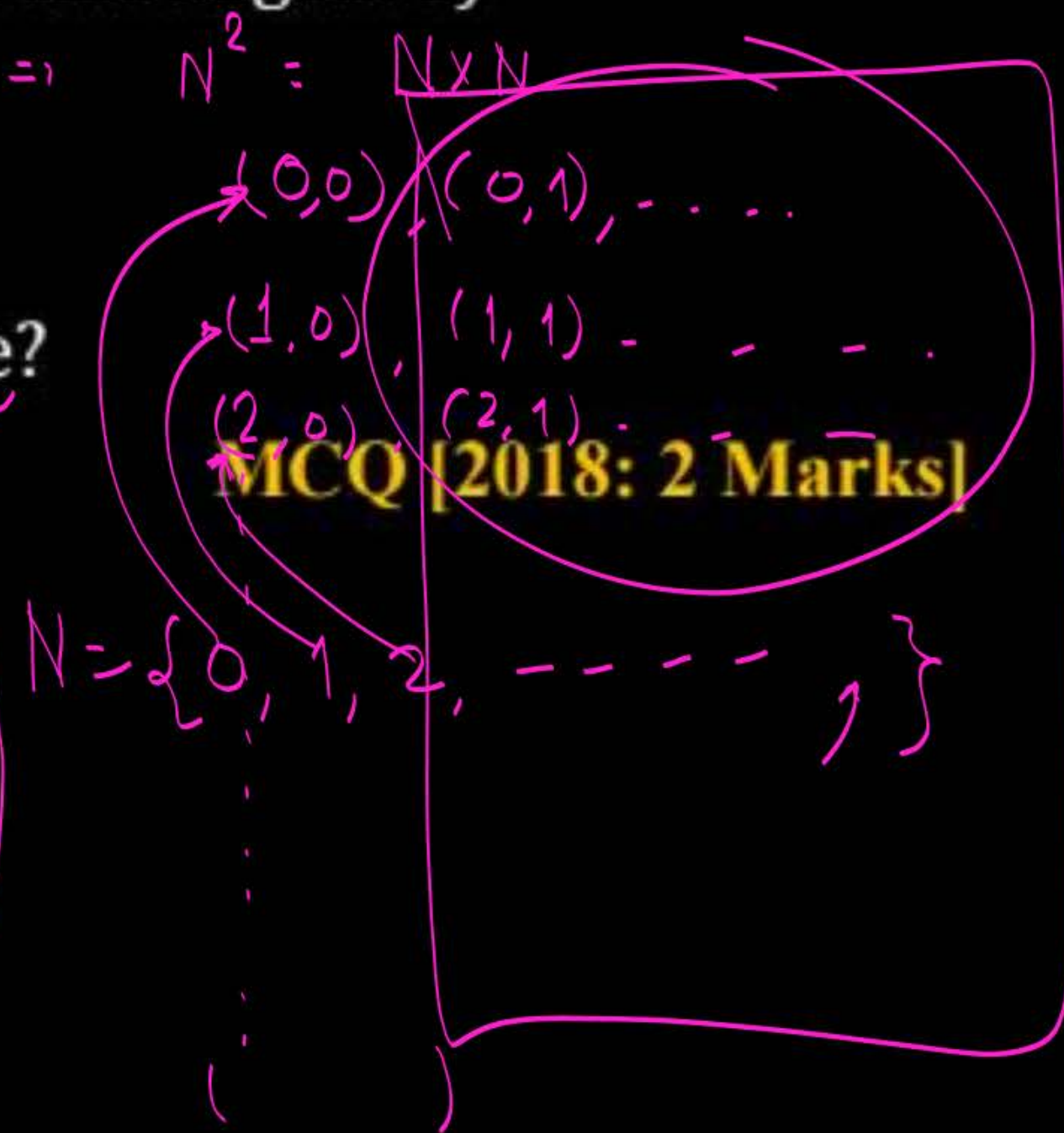
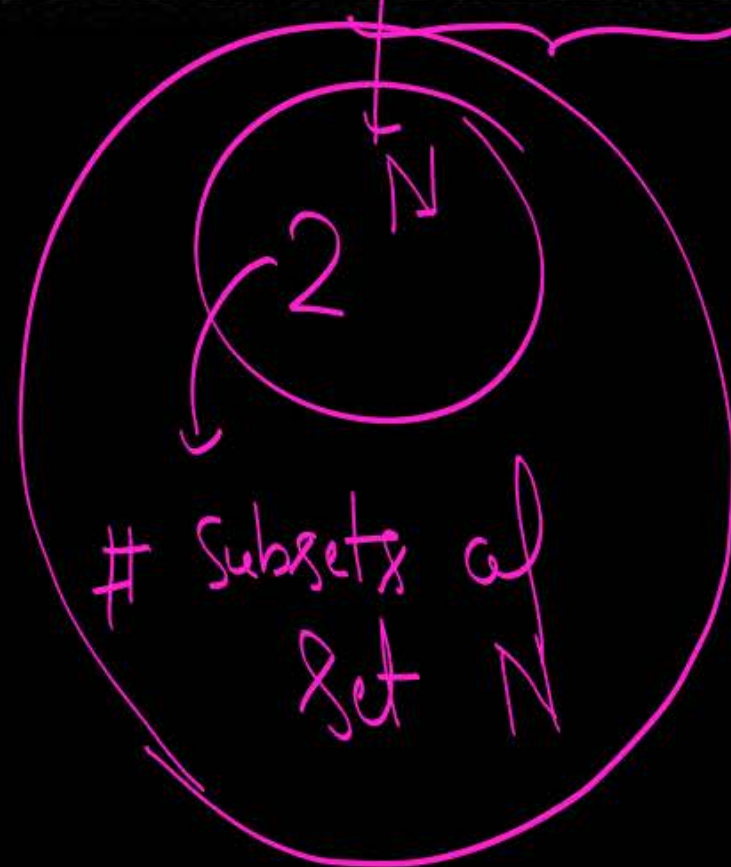


Let N be the set of natural numbers. Consider the following sets

- ☒ P: Set of Rational numbers (positive and negative)
- ☒ Q: Set of functions from $\{0, 1\}$ to N .
- ☒ R: Set of functions from N to $\{0, 1\}$.
- ☒ S: Set of finite subsets of N .

Which of the sets above are countable?

- ☐ A Q and S only
- ☒ B P and S only
- ☐ C P and R only
- ☐ D P, Q and S only



Consider the following sets, where $n \geq 2$:

S_1 : Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$

S_2 : Set of all functions from the set $\{0, 1, 2, \dots, n^2 - 1\}$ to the set $\{0, 1, 2\}$

Which of the following choice(s) is/are correct?

There exists a surjection from S_1 to S_2 **MSQ [2021 (Set-2): 1 Mark]**

There does not exist a bijection from S_1 to S_2 .

There does not exist an injection from S_1 to S_2 .

There exists a bijection from S_1 to S_2 .



A pennant is a sequence of numbers, each number being 1 or 2. An n -pennant is a sequence of numbers with sum equal to n . For example, $(1,1,2)$ is a 4-pennant. The set of all possible 1-pennants is $\{(1)\}$, the set of all possible 2-pennants is $\{(2), (1,1)\}$ and the set of all 3-pennants is $\{(2,1), (1,1,1), (1,2)\}$.

Note that the pennant $(1,2)$ is not the same as the pennant $(2,1)$.

The number of 10-pennants is _____. **NAT [2014 (Set-1): 2 Marks]**

Let $U = \{1, 2, \dots, n\}$ and $A = \{x, X\}$, $x \in X$ and $X \subseteq U$. Consider the following two statements on $|A|$.

I: $|A| = n \cdot 2^{n-1}$

II: $|A| = \sum_{k=1}^n k \cdot \binom{n}{k}$

Which of the above statements is/are correct?

MCQ [2019: 1 Mark]

A

Both I and II

B

Neither I nor II

C

Only II

D

Only I

(x, X)

if $X = \emptyset$
then

$(_, \emptyset)$

no element
belongs to $\emptyset$

$\therefore$ Such order pair not possible
Hence X can not be $\emptyset$



Let $U = \{1, 2, \dots, n\}$ and $A = \{(x, X), x \in X \text{ and } X \subseteq U\}$. Consider the following two statements on $|A|$.

I: $|A| = n \cdot 2^{n-1}$

II: $|A| = \sum_{k=1}^n k \cdot \binom{n}{k}$

Which of the above statements is/are correct?

MCQ [2019: 1 Mark]

A

Both I and II

B

Neither I nor II

C

Only II

D

Only I

$$\begin{array}{ccccccc} (x, X) & \text{or} & (x, X) & \text{or} & (x, X) & \dots & (x, X) \\ \downarrow |X|=1 & & \downarrow |X|=2 & & \downarrow |X|=3 & & \downarrow |X|=n \end{array}$$

$$|A| = \binom{1}{1} + \binom{2}{2} + \binom{3}{3} + \dots + \binom{n}{n}$$

$$\begin{array}{l} (1, \{1\}) \\ (2, \{2\}) \end{array}$$

$$= \sum_{k=1}^n k \cdot \binom{n}{k}$$

Let $U = \{1, 2, \dots, n\}$ and $A = \{(x, X), x \in X \text{ and } X \subseteq U\}$. Consider the following two statements on $|A|$.

I: $|A| = n \cdot 2^{n-1}$

II: $|A| = \sum_{k=1}^n k \cdot \binom{n}{k}$

Which of the above statements is/are correct?

MCQ [2019: 1 Mark]

☒ A

Both I and II

☐ B

Neither I nor II

☐ C

Only II

☐ D

Only I

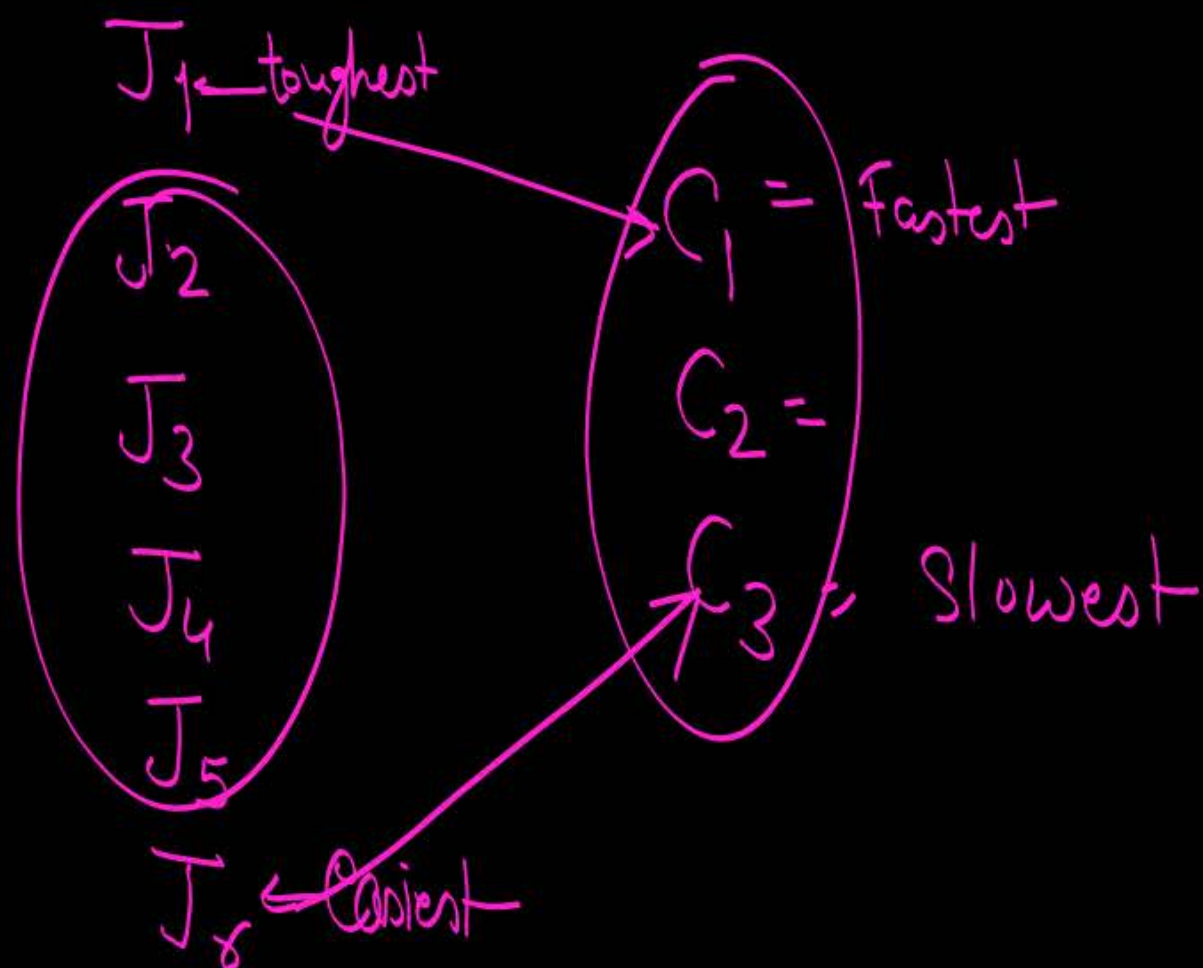
(x, X)
 $|A| = (n \cdot 2^{n-1})$

There are 6 jobs with distinct difficulty levels, and 3 computers with distinct processing speeds. Each job is assigned to a computer such that:

- The fastest computer gets the toughest job and the slowest computer gets the easiest job.
- ~~Every computer gets at least one job.~~

The number of ways in which this can be done is _____.

NAT [2021 (Set-1): 1 Mark]



$$\begin{aligned}
 & (3)^4 - (2)^4 \\
 &= 81 - 16 \\
 &= \boxed{65}
 \end{aligned}$$

Q.

Sets, Relations and Functions



1. [MSQ]

[GATE-2023: 2M]

Let $f : A \rightarrow B$ be an onto (or surjective) function, where A and B are nonempty sets. Define an equivalence relation $\sim$ on the set A as

$a_1 R a_2$ iff $f(a_1) = f(a_2)$
 $a_1 \sim a_2$ if $f(a_1) = f(a_2)$,

Symmetric ✓
 Reflexive ✓
 Transitive ✓

Where $a_1, a_2 \in A$. Let $\varepsilon = \{[x] : x \in A\}$ be the set of all the equivalence classes under $\sim$. Define a new mapping $F : \varepsilon \rightarrow B$ as

$F([x]) = f(x)$, for all the equivalence classes $[x]$ in ε .

Which of the following statements is/are TRUE?

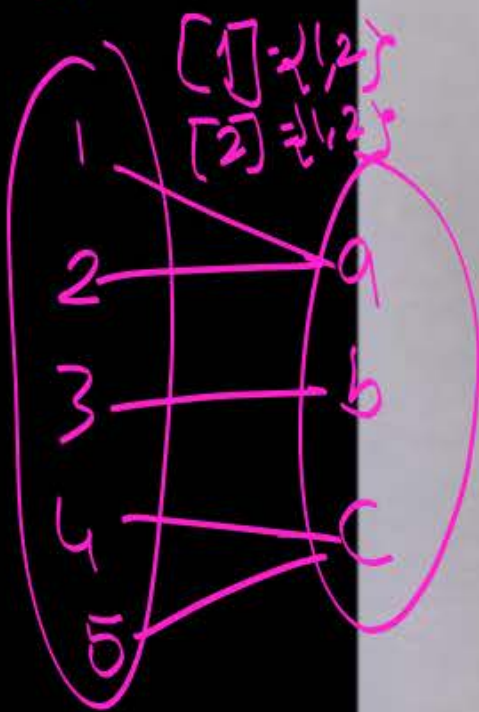
☒ (a) F is NOT well – defined.

☒ (b) F is an onto (or surjective) function.

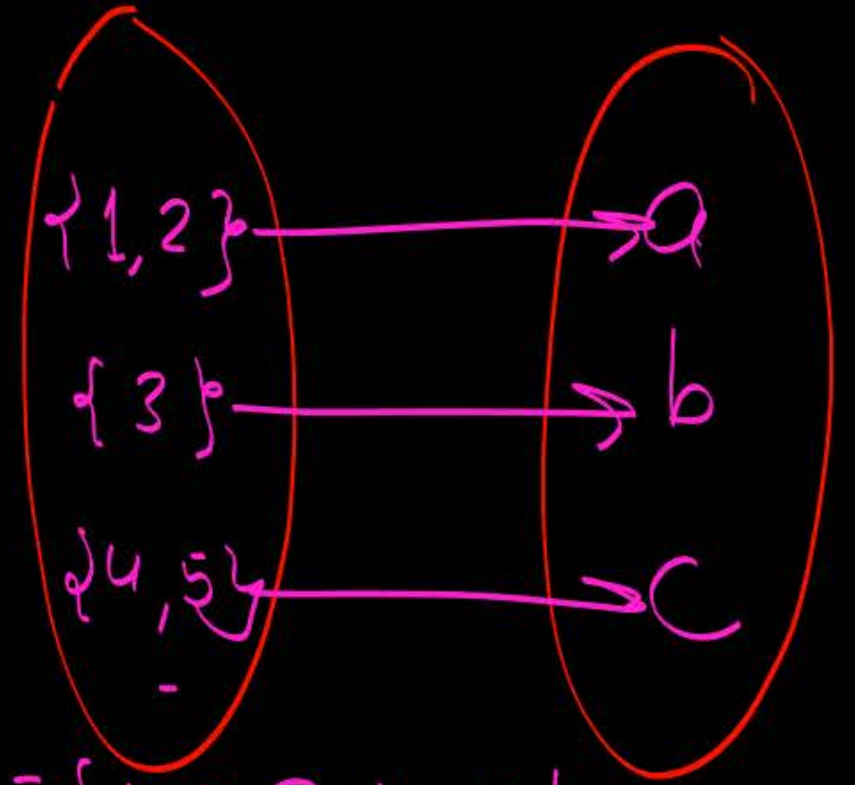
☒ (c) F is a one-to-one (or injective) function.

☒ (d) F is a bijective function.

$f: A \rightarrow B$



Partition of A w.r.t. Equivalence relation $\sim$
 $\rightarrow F: \varepsilon \rightarrow B$



$A = \{1, 2, 3, 4, 5\}$
 $\varepsilon = \{\{1, 2\}, \{3\}, \{4, 5\}\}$

H.W.



Q.42	Let $\mathcal{F}$ be the set of all functions from $\{1, \dots, n\}$ to $\{0, 1\}$. Define the binary relation $\preceq$ on $\mathcal{F}$ as follows: $\forall f, g \in \mathcal{F}, f \preceq g$ if and only if $\forall x \in \{1, \dots, n\}, f(x) \leq g(x)$, where $0 \leq 1$. Which of the following statement(s) is/are TRUE?
(A)	$\preceq$ is a symmetric relation
(B)	$(\mathcal{F}, \preceq)$ is a partial order
(C)	$(\mathcal{F}, \preceq)$ is a lattice
(D)	$\preceq$ is an equivalence relation

① Reflexive:

Let $f \in F$

$$f(1) \leq^{\checkmark} f(1) \quad \left\{ \begin{array}{l} \text{because} \\ f(1) = f(1) \end{array} \right\}$$

$$f(2) \leq f(2)$$

$$f(3) \leq f(3)$$

$\vdots$

$$f(n) \leq f(n)$$

$$\forall x \in \{1, \dots, n\} \quad f(x) \leq f(x)$$

$\therefore \underline{f R f}$ is Reflexive

Symmetric:

Let $f, g \in F$ & $f \neq g$

$$\text{let } f(x) = 0 \quad \forall x \in \{1, 2, \dots, n\}$$

$$\& \quad g(x) = 1 \quad \forall x \in \{1, \dots, n\}$$

$$f(x) \leq g(x) \quad \forall x \in \{1, \dots, n\}$$

$$\therefore f R g \quad \text{but} \quad g \not R f$$

$\therefore$ Not Symmetric

Anti-symmetric:

Let $f, g \in F$
And assume $f R g$
i.e. $f(1) \leq g(1)$
 $f(2) \leq g(2)$
 $\vdots$
 $f(n) \leq g(n)$

} if $f \neq g$
then for at least one element 'k' in set
 $\{1, 2, \dots, n\}$
 $f(k) < g(k)$, then $g(k) \neq f(k)$

$\downarrow$
Hence $g(x) \leq f(x)$
 $\forall x \in \{1, \dots, n\}$

$\therefore g R f$ Hence Anti-symmetric

There exist a function $M \in F$

$$\text{s.t. } M(x) = 1, \quad \forall x \in \{1, \dots, n\}$$

then $M(x)$ will be the maximum element
of POSET

There exist a function $N \in F$

$$\text{s.t. } N(x) = 0, \quad \forall x \in \{1, \dots, n\}$$

then $N(x)$ will be the minimum element
of POSET

★

Let $f, g \in F$
 s.t. $f \neq g$
 and $f \leq g$ i.e. $f(x) \leq g(x) \forall x \in \{1, \dots, n\}$

$f(x_1) = 0$	$f(x_2) = 0$	$f(x_3) = 0$
$g(x_1) = 0$	$g(x_2) = 0$	$g(x_3) = 1$

$$\lambda(k) = \sigma(k) =$$

$$\begin{cases} \lambda(x_1) = 1 & \lambda(x_2) = 0 & \lambda(x_3) = 1 \\ \sigma(x_1) = 0 & \sigma(x_2) = 1 & \sigma(x_3) = 1 \end{cases}$$

$$\lambda(x_1) = 1 \neq \sigma(x_1) = 0$$

Can we have two functions $\lambda, \sigma \in F$

$$s.t. f \leq \lambda \neq f \leq \sigma$$

$$g \leq \lambda \neq g \leq \sigma$$

$$f(x_1) = 0 \neq f(x_2) = 0$$

$\lambda \neq \sigma$ are
not comparable

$$g(x_1) = 0 \neq g(x_2) = 0$$

$$'k' \quad f(k) < g(k)$$

$$\begin{cases} \text{for some } x_1 \\ \lambda(x_1) \neq \sigma(x_1) \\ \text{for some } x_2 \\ \sigma(x_2) \neq \lambda(x_2) \end{cases}$$

$$\sigma(x_2) = 1 \neq \lambda(x_2) = 0$$

Q.17

$g(.)$ is a function from A to B , $f(.)$ is a function from B to C , and their composition defined as $f(g(.))$ is a mapping from A to C .

If $f(.)$ and $f(g(.))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(.)$?

~~(A)~~

$g(.)$ must be an onto (surjective) function.

~~(B)~~

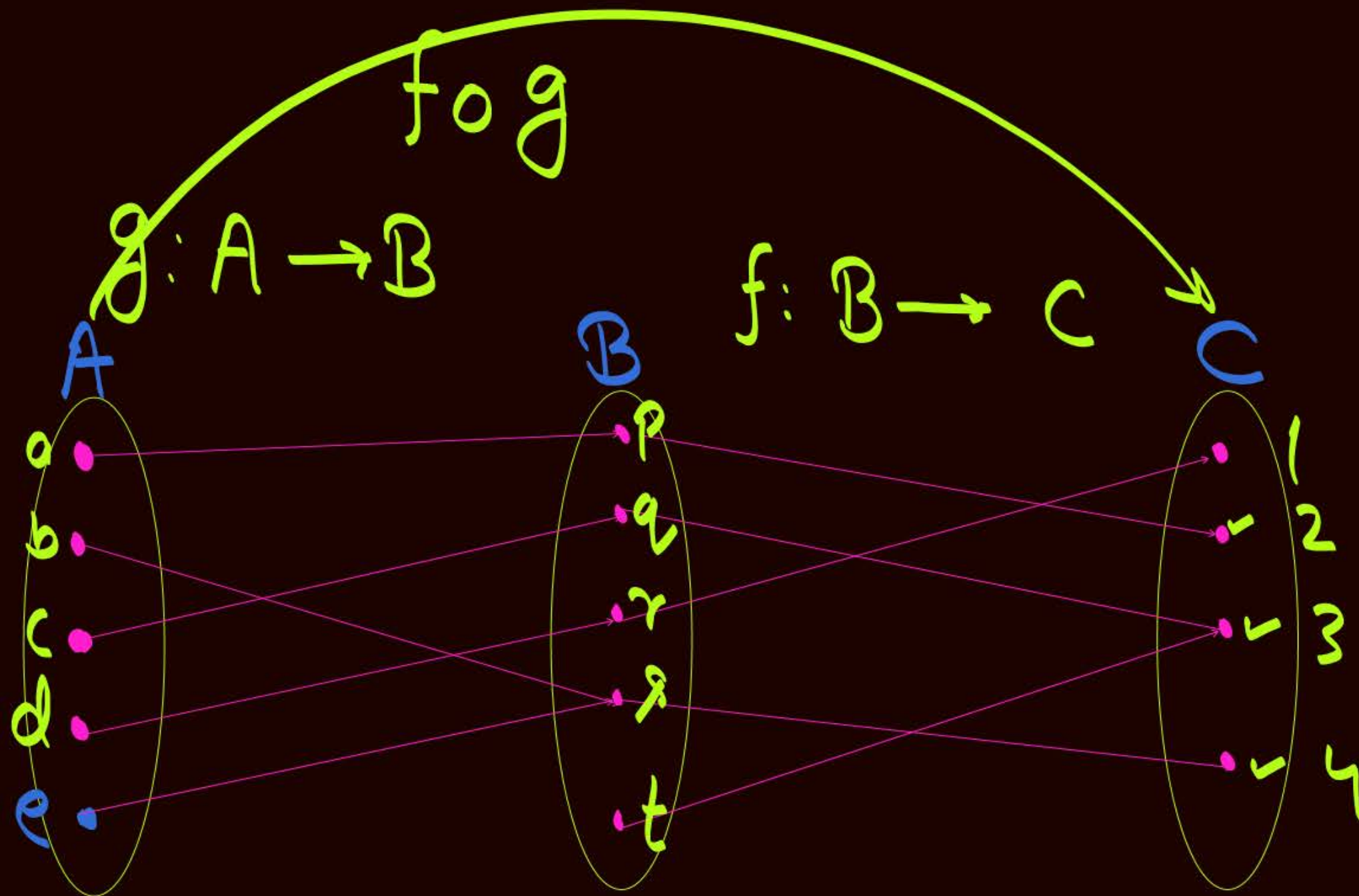
$g(.)$ must be a one-to-one (injective) function.

~~(C)~~

$g(.)$ must be a bijective function, that is, both one-to-one and onto.

☒ (D)

$g(.)$ is not required to be a one-to-one or onto function.





2 mins Summary



Topic

Practice questions & PYQ

Slide

THANK - YOU